

ISHA GUPTA

📞 +91 8968454168 ✉️ guptaishaa09@gmail.com 🔗 linkedin.com/in/isha-gupta/ 🌐 github.com/Ishaaguptaaa

EDUCATION

Thapar Institute of Engineering & Technology

Bachelor of Engineering - Computer Science and Engineering

July 2023 – June 2027

CGPA: 9

EXPERIENCE

Experiential Learning Centre-TIET

Summer Intern

June 2024 – August 2024

- Programmed and debugged **Arduino based motion control** for a cleaning robot, implementing navigation and obstacle detection using **ultrasonic/IR sensors**.
- Integrated **Arduino, motor drivers, and sensor modules** to fine tune robot movement and improve system performance.

Mehta Family School for Data Science and Artificial Intelligence - IIT Roorkee

Summer Intern

June 2025 – July 2025

- Researched **Differential Privacy** mechanisms and analyzed privacy vulnerabilities in machine learning systems, including **membership inference attacks**.
- Implemented Python based simulations of **privacy attacks** and performed data analysis to evaluate the robustness of differential privacy techniques.

Samsung PRISM

Research Intern

October 2025 – Feb 2026

- Developed machine learning models for enhancement assessment in **RAW DNG image buffers** and built an end to end **ETL pipeline** for data processing and evaluation.
- Built a **Python pipeline** for DNG decoding, RAW buffer extraction, image enhancement, and quantitative quality assessment.
- Researched and applied **image quality metrics** to evaluate enhancement performance and validate model outputs.

TECHNICAL SKILLS

Languages: Python | C++ | C | SQL | R

Machine Learning: Deep Learning | Computer Vision | Generative AI | LLMs | Multimodal Learning | Privacy Preserving ML

Frameworks: PyTorch | TensorFlow | Scikit Learn | OpenCV | Hugging Face | YOLO

Web Development: HTML | CSS | JavaScript | React.js | Node.js | Express.js | REST API

Databases: MySQL | PostgreSQL | MongoDB

Tools: Git | GitHub | Postman | Jupyter Notebook | MATLAB | VS Code

Core CS: Data Structures & Algorithms | OOPs | DBMS | Operating Systems

Data Engineering: ETL Pipelines | Data Cleaning | Data Transformation | Data Analysis

PROJECTS

PrivaScanFL: Federated Brain MRI Classification

- Developed a **deep learning** pipeline for multi class brain MRI tumor classification using federated learning.
- Implemented and benchmarked CNN architectures including **ResNet50, DenseNet121, MobileNetV2, and VGG16** to identify the best performing baseline model for federated training.
- Applied federated learning algorithms including **FedAvg, FedProx, FedNova, and FedMOON** for decentralized model training across multiple clients.
- Performed **Membership Inference Attacks (MIA)** to evaluate privacy leakage and analyze the robustness of federated learning models.
- Technologies Used: **Python, PyTorch/TensorFlow, Federated Learning, Deep Learning, Differential Privacy**.

AgniPathik: AI Powered Smart Fire Evacuation System

- Developed an **AI-driven fire evacuation system** for dynamic and congestion aware emergency route planning.
- Implemented **machine learning models** for fire risk analysis and crowd congestion prediction using real time environmental and occupancy data.
- Contributed to **fire simulation**, dynamic path planning, and system testing under changing fire conditions.
- Technologies Used: **Python, Machine Learning, XGBoost, Random Forest, IoT, ESP32, Path Planning Algorithms**.

ACHIEVEMENTS

- Awarded **Undergraduate Research Fellowship (URF)** worth **20K** for the research project titled “**Privacy Preserving Brain Tumor Classification using Federated Learning on Distributed MRI Data**” by TIET, patiala.
- Research paper on **privacy preserving neuroimaging analysis** accepted at a conference organized by **Sejong University, South Korea** and selected for **Springer LNNS** publication.
- Selected among the **Top 100 Teams** in the **ZS BEATS Challenge**.
- **Winner of Makeathon'24: An Overnight Hackathon**, organized by **TIET Patiala**.
- Recognized among the **Top 10 Projects** for Autonomous Cleaning Robot during Summer Internship organized by **Experiential Learning Centre'24, TIET**.